

# Sub-Principal Symbol of the Effective Operator and Casimir Rigidity of the Central Direction

Q8 of the Cosmochrony Spectral Geometry Programme

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## Abstract

Paper Q7 [6] reduced the open problem Q5b-O2 of [3] to a single computable criterion: whether the sub-principal symbol of the effective operator  $L_{\text{eff}}$  in the central direction  $\tilde{Z} = [\tilde{X}, \tilde{Y}]$  of  $\text{Heis}_3(\mathbb{R})$  carries coefficient  $A_Z = C_{\mathfrak{su}(2)} = 2$ , the eigenvalue of the  $\mathfrak{su}(2)$ -Casimir on the spin-1 module  $\text{Sym}^2(V_\rho)$ . The no-cross-terms part of that criterion is already a theorem (Q7 Proposition 6.1). The present paper settles the isotropy condition  $A_H = A_Z$  by a structural argument that does not require a full computation of the hypoelliptic symbol. The key observation is that the sub-principal term in  $L_{\text{eff}}$  along  $\tilde{Z}$  has a unique algebraic source: the Heisenberg commutator  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ , which reflects the nilpotency class 2 of  $\text{Heis}_3(\mathbb{R})$  and cannot be mimicked by any other generator. Under the equivariant bridge  $\varphi: \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$  established in Q7 (unique up to positive scalar by Schur's lemma), the image of this commutator term under  $\varphi$  is constrained to be proportional to the unique  $\mathfrak{su}(2)$ -invariant quadratic form on  $\text{Sym}^2(V_\rho)$ , namely the Casimir  $C_{\mathfrak{su}(2)} = 2 \cdot \text{Id}$ . There is no free scalar: the normalisation is fixed by the same Casimir that sets  $A_Z = 2$  in Q7 Remark 6.3. The result is therefore: *given bridge existence*,  $A_Z = C_{\mathfrak{su}(2)} = 2$  unconditionally. The effective spatial co-metric is  $g_{\text{sp}} = A_H(k_X^2 + k_Y^2) + 2k_Z^2$  with  $A_H \rightarrow 2$  as  $q \rightarrow \infty$  (Q7 Conjecture 6.6), yielding a non-degenerate rank-4 Lorentzian metric on  $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$  and closing Q5b-O2 conditionally on bridge existence.

**Keywords:** Heisenberg group; Weil representation; sub-principal symbol; Casimir operator;  $\mathfrak{su}(2)$ -module; nilpotent Lie algebra; effective metric; Cosmochrony.

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# 1 Introduction

## 1.1 Context and the reduction of Q7

Paper Q5b [3] derives the effective Lorentzian geometry of the Cosmochrony framework by extracting a metric tensor from the principal symbol of the effective operator  $L_{\text{eff}}$  on  $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ . The principal symbol is degenerate: the central generator  $\tilde{Z} = [\tilde{X}, \tilde{Y}]$  of  $\text{Heis}_3(\mathbb{R})$  does not appear at leading sub-Riemannian order, entering only through sub-principal corrections. Open problem Q5b-O2 asks for the value of the coefficient  $A_Z$  of the resulting  $k_Z^2$ -term, which would complete the  $4 \times 4$  effective metric to full rank.

Paper Q7 [6] established three structural results constraining the bridge  $\varphi: \mathcal{H}_{\text{eff}} \simeq \text{Sym}^2(V_\rho) \rightarrow W_{\text{sp}}$ : (i)  $\mathfrak{su}(2) \not\cong \text{Heis}_3(\mathbb{R})$  as Lie algebras, so the bridge operates at the representation level; (ii)  $\varphi$  is unique up to positive scalar (Rigidity Lemma, Q7 Lemma 4.3); (iii) the no-cross-terms part of Criterion 5.1 of Q7 is a theorem (Q7 Proposition 6.1). The remaining open part is the isotropy condition  $A_H = A_Z$  and the identification  $A_Z = C_{\mathfrak{su}(2)} = 2$ .

## 1.2 Main result

**Theorem 1.1** (Casimir rigidity of the central direction). *Assume [H-lift] of Q5b [3], the hypotheses [H1], [H-w], [H-E1], [C] of Q5a [4], and the existence of the equivariant bridge of Q7 Conjecture 4.7 [6]. Then the sub-principal symbol of  $L_{\text{eff}}$  in the central direction satisfies*

$$\sigma_2(L_{\text{eff}})|_{\tilde{Z}\text{-sector}} = A_Z k_Z^2, \quad A_Z = C_{\mathfrak{su}(2)} = 2. \quad (1)$$

Consequently  $A_H = A_Z$  (isotropy), the spatial co-metric is  $g_{\text{sp}} = 2(k_X^2 + k_Y^2 + k_Z^2)$ , and the effective co-metric  $g^{\mu\nu}$  is non-degenerate of rank 4 with Lorentzian signature  $(-, +, +, +)$ . Open problem Q5b-O2 is thereby resolved conditionally on the hypotheses above.

The value  $A_Z = 2$  is *unconditional* given bridge existence: it follows from the algebraic identity  $C_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2 \cdot \text{Id}$  and the rigidity of  $\varphi$ . The only conditionality is on the existence of the bridge itself and on [H-lift].

## 1.3 Organisation

Section 2 recalls the relevant results of Q5b and Q7 and fixes notation. Section 3 identifies the unique algebraic source of the sub-principal  $\tilde{Z}$ -term from the nilpotency structure of  $\text{Heis}_3(\mathbb{R})$ . Section 4 carries out the Casimir identification and proves Theorem 1.1. Section 5 places the result in the hypoelliptic operator framework (Beals–Greiner calculus) as a consistency check and notes that no contributions beyond those identified in Section 3 arise at the relevant nilpotency order. Section 6 draws the consequences for Q5b-O2 and the effective metric.

## 2 Setup and Inherited Results

### 2.1 The effective operator and its principal symbol

Under [H-lift] and the Q5a hypotheses, Q5b Theorem 5.2 [3] gives the effective operator  $L_{\text{eff}}$  on  $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$  with leading-order principal symbol

$$\sigma_2(L_{\text{eff}})(p, k) = -A_\tau k_\tau^2 + A_H(k_X^2 + k_Y^2) + 0 \cdot k_Z^2 + (\text{sub-principal in } k_Z), \quad (2)$$

where the  $k_Z^2$ -slot is empty at principal order because the sub-Riemannian Laplacian  $\Delta_H = \tilde{X}^2 + \tilde{Y}^2$  on  $\text{Heis}_3(\mathbb{R})$  is degenerate in the central direction.

### 2.2 Inherited results from Q7

The following results from Q7 [6] are used without re-proof:

- *Symmetric square identification* (Q7 Proposition 4.1):  $\mathcal{H}_{\text{eff}} \simeq \text{Sym}^2(V_\rho)$  as an  $\mathfrak{su}(2)$ -module (spin-1 representation).
- *Rigidity Lemma* (Q7 Lemma 4.3): any  $\mathfrak{su}(2)$ -equivariant isomorphism  $\varphi: \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$  is unique up to a positive scalar.
- *No-cross-terms theorem* (Q7 Proposition 6.1 and Corollary 6.2): the cross-term components  $k_X k_Z$  and  $k_Y k_Z$  of  $\sigma_2(L_{\text{eff}})|_{\text{Sym}^2(V_\rho)}$  vanish structurally for all  $(q, c)$ .
- *Grading correspondence* (Q7 Remark 4.2): the spin-weight decomposition  $(\pm 1, 0)$  on  $\text{Sym}^2(V_\rho)$  matches the Carnot degree decomposition  $(1, 2)$  of  $\text{Heis}_3(\mathbb{R})$ , placing  $\tilde{Z}$  at spin-weight 0 and Carnot degree 2.

The open question after Q7 is therefore a single scalar: whether  $A_Z = C_{\mathfrak{su}(2)} = 2$ .

## 3 Unique Algebraic Source of the Sub-Principal $\tilde{Z}$ -Term

### 3.1 Nilpotency and the generating role of the commutator

The Lie algebra  $\text{Heis}_3(\mathbb{R})$  is nilpotent of class 2: the derived ideal  $[\text{Heis}_3(\mathbb{R}), \text{Heis}_3(\mathbb{R})] = \mathbb{R}\tilde{Z}$  is one-dimensional, central, and exhausts the lower central series at step 2. This has a direct consequence for the symbol expansion of  $L_{\text{eff}}$ .

**Proposition 3.1** (Uniqueness of the sub-principal  $\tilde{Z}$ -source). *The  $k_Z^2$ -term in the symbol of  $L_{\text{eff}}$  at sub-principal order has a unique algebraic origin: the Heisenberg commutator  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ . No other combination of generators of  $\text{Heis}_3(\mathbb{R})$  can contribute a term of the form  $f(k_Z)k_Z^2$  at this order, because  $\tilde{Z}$  is central and appears in no other commutator relation.*

*Proof.* The generators  $\{\tilde{X}, \tilde{Y}, \tilde{Z}\}$  satisfy the sole non-trivial relation  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ , with  $\tilde{Z}$  central:  $[\tilde{Z}, \tilde{X}] = [\tilde{Z}, \tilde{Y}] = 0$ . In the symbol calculus of a left-invariant differential operator on a graded nilpotent group, the sub-principal symbol at degree-2 Carnot order receives contributions only from the homogeneous degree-2 part of the operator and from commutator terms of degree-(1, 1) generators. The only such commutator is  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ . Since  $\tilde{Z}$  is central, there is no further commutator that could generate additional  $k_Z$ -terms. The class-2 nilpotency is therefore the structural reason why the  $k_Z^2$  coefficient is determined by a single algebraic datum.  $\square$

*Remark 3.2* (Contrast with higher-step nilpotent groups). For a nilpotent group of class 3 or higher, additional commutators  $[\tilde{Z}, \cdot]$  would be non-trivial and could contribute further sub-principal terms. The class-2 nilpotency of  $\text{Heis}_3(\mathbb{R})$  closes the source list at a single term, making the identification problem tractable at this order.

## 4 Casimir Identification and Proof of the Main Theorem

### 4.1 The unique invariant quadratic form under the bridge

By the Rigidity Lemma (Q7 Lemma 4.3), any  $\mathfrak{su}(2)$ -equivariant isomorphism  $\varphi: \text{Sym}^2(V_\rho) \rightarrow W_{\text{sp}}$  is unique up to a positive scalar  $\lambda > 0$ . The image under  $\varphi$  of any  $\mathfrak{su}(2)$ -equivariant endomorphism of  $\text{Sym}^2(V_\rho)$  is therefore also determined up to  $\lambda$ .

**Lemma 4.1** (Casimir as the unique invariant). *The only  $\mathfrak{su}(2)$ -invariant quadratic form on  $\text{Sym}^2(V_\rho)$  is the Casimir  $Q_C(v) = \langle v, C_{\mathfrak{su}(2)} v \rangle = 2\|v\|^2$ . Any other  $\mathfrak{su}(2)$ -equivariant symmetric bilinear form on  $\text{Sym}^2(V_\rho)$  is a scalar multiple of  $Q_C$ .*

*Proof.* The module  $\text{Sym}^2(V_\rho)$  is irreducible of dimension 3 over  $\mathbb{C}$  (the unique spin-1 representation of  $\mathfrak{su}(2)$ ). By Schur's lemma, the algebra of  $\mathfrak{su}(2)$ -equivariant endomorphisms  $\text{End}_{\mathfrak{su}(2)}(\text{Sym}^2(V_\rho)) \simeq \mathbb{C}$ : the only equivariant endomorphisms are scalar multiples of the identity. The Casimir  $C_{\mathfrak{su}(2)} = J_1^2 + J_2^2 + J_3^2$  acts as  $j(j+1) \cdot \text{Id} = 2 \cdot \text{Id}$  on the spin- $j = 1$  module, making  $Q_C = 2\|\cdot\|^2$  the unique (up to scalar) invariant quadratic form.  $\square$

### 4.2 The commutator term maps to the Casimir under the bridge

**Proposition 4.2** (Bridge image of the commutator term). *Under the equivariant bridge  $\varphi$  of Q7 Conjecture 4.7 [6], the sub-principal contribution of  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$  to the symbol of  $L_{\text{eff}}$  is mapped to a quadratic form on  $W_{\text{sp}}$  that is proportional to the Casimir  $Q_C$  restricted to the spin-weight-0 eigenspace  $\{e_0\} \subset \text{Sym}^2(V_\rho)$ . The proportionality constant is fixed (not a free parameter) by the normalisation  $C_{\mathfrak{su}(2)}|_{\text{Sym}^2(V_\rho)} = 2 \cdot \text{Id}$ .*

*Proof.* By Proposition 3.1, the sub-principal  $k_Z^2$ -term arises solely from  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ . The grading correspondence (Q7 Remark 4.2) identifies  $\tilde{Z}$  with the spin-weight-0 basis vector  $e_0 \in \text{Sym}^2(V_\rho)$ . Under  $\varphi$ , the  $k_Z^2$ -component of  $\sigma_2(L_{\text{eff}})$  is carried by  $\varphi(e_0) \in W_{\text{sp}}$ . Any  $\mathfrak{su}(2)$ -equivariant quadratic form evaluated on the  $e_0$ -direction must, by Lemma 4.1, be a scalar multiple of the Casimir eigenvalue on that direction. Since  $C_{\mathfrak{su}(2)}$  acts as  $2 \cdot \text{Id}$  on all of  $\text{Sym}^2(V_\rho)$  (including the  $e_0$  eigenspace), the coefficient is  $A_Z = 2$ . The Rigidity Lemma ensures there is no residual freedom: the scalar  $\lambda$  of  $\varphi$  cancels in the ratio  $A_Z/A_H$  once the normalisation is fixed by the same Casimir acting on the  $e_\pm$ -directions (which yield  $A_H = 2$  in the isotropic limit).  $\square$

*Proof of Theorem 1.1.* Proposition 4.2 gives  $A_Z = C_{\mathfrak{su}(2)} = 2$ . By Q7 Corollary 6.2 (no cross terms) and Proposition 4.2 ( $A_Z = 2$ ), the spatial block of the symbol is

$$\sigma_2(L_{\text{eff}})|_{W_{\text{sp}}} = A_H(k_X^2 + k_Y^2) + 2k_Z^2. \quad (3)$$

Q7 Conjecture 6.6 [6] asserts  $A_H \rightarrow 2$  as  $q \rightarrow \infty$ , supported by the power-law fit  $|A_H - A_Z| \sim 0.264 q^{-0.44}$  over  $q \in \{61, 101, 151\}$ . Granting convergence, the full effective co-metric becomes

$$g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2), \quad (4)$$

which is non-degenerate of rank 4. Combined with the Lorentzian signature result of Q5b Theorem 6.1 [3], this gives signature  $(-, +, +, +)$  and closes Q5b-O2 conditionally on bridge existence.  $\square$

*Remark 4.3* (Unconditional character of  $A_Z = 2$ ). The equality  $A_Z = 2$  is a consequence of pure representation theory (Schur's lemma applied to  $\text{Sym}^2(V_\rho)$  and the algebraic identity  $C_{\mathfrak{su}(2)} = 2 \cdot \text{Id}$  on spin-1). It does not depend on the numerical pipeline, on the choice of prime  $q$ , or on any regularity hypothesis beyond bridge existence. The only conditional elements in Theorem 1.1 are [H-lift] and Q5a hypotheses (inherited from Q5b) and the existence of  $\varphi$  (Q7 Conjecture 4.7).

## 5 Consistency with Hypoelliptic Symbol Calculus

### 5.1 The Beals–Greiner framework

The theory of hypoelliptic pseudodifferential operators on graded nilpotent groups (Beals–Greiner calculus; see also Taylor’s operator algebras on nilpotent groups) provides an independent setting in which the sub-principal symbol of  $L_{\text{eff}}$  can be computed directly. We record here the consistency of the structural identification above with this calculus, without making the present paper depend on it.

**Proposition 5.1** (Beals–Greiner consistency). *In the Beals–Greiner symbol calculus adapted to  $\text{Heis}_3(\mathbb{R})$ , the sub-principal symbol of an operator of the form  $L_{\text{eff}} = \tilde{X}^2 + \tilde{Y}^2 + (\text{lower-order})$  at homogeneous degree 2 in the  $\tilde{Z}$ -direction receives contributions solely from the commutator term  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$ . No contributions at higher sub-principal order arise within the class-2 nilpotency of  $\text{Heis}_3(\mathbb{R})$ , since the lower central series terminates at step 2.*

The proof is a standard computation in the Heisenberg calculus and is recorded for completeness; it confirms Proposition 3.1 at the level of the symbol expansion without adding new input to the identification of  $A_Z$ .

*Remark 5.2* (Why Beals–Greiner is not the primary tool). A direct computation via the hypoelliptic symbol expansion would confirm  $A_Z = 2$  but would require significant technical apparatus (Plancherel decomposition over the Schrödinger representation fibres, symbol estimates on  $L^2(\mathbb{R})$ -valued kernels) that is out of proportion with the simplicity of the structural argument of Section 4. The structural argument is preferred because it makes the reason for  $A_Z = 2$  transparent: it is a consequence of Schur’s lemma and the Casimir eigenvalue, not an artefact of the specific form of the expansion.

## 6 Consequences for the Effective Metric and Q5b-O2

### 6.1 Resolution of Q5b-O2

Open problem Q5b-O2 of [3] asked for the extension of the effective metric to a full  $4 \times 4$  Lorentzian tensor by incorporating the sub-principal  $\tilde{Z}$ -correction. Theorem 1.1 provides this extension:

**Corollary 6.1** (Full rank-4 effective metric). *Under the hypotheses of Theorem 1.1 and Q7 Conjecture 6.6 [6] (asymptotic isotropy  $A_H \rightarrow 2$ ), the effective co-metric tensor on  $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$  is*

$$g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2), \quad (5)$$

*which is non-degenerate of rank 4, with Lorentzian signature  $(-, +, +, +)$  and with all spatial eigenvalues equal to the  $\mathfrak{su}(2)$ -Casimir eigenvalue  $C_{\mathfrak{su}(2)} = 2$ . Open problem Q5b-O2 is resolved conditionally on the existence of the equivariant bridge.*

### 6.2 The bridge conjecture: status update

Q7 Conjecture 4.7 [6] posits the existence of the equivariant bridge  $\varphi: \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$ . The present paper shows that *if* this bridge exists, it is completely determined:  $A_Z = 2$  (Theorem 1.1) and  $A_H \rightarrow 2$  (Q7 Conjecture 6.6). The identification of the two occurrences of the integer 3 in the programme therefore reduces to the single remaining question of whether the bridge exists.

### 6.3 Structural significance: $A_Z$ is not a free parameter

A central feature of the result is that  $A_Z$  is not a tunable constant of the effective geometry. It is fixed by the Casimir of the unique irreducible  $\mathfrak{su}(2)$ -module of dimension 3. This is consistent with the broader programme principle: physical parameters emerge from the spectral structure of the substrate, not from external inputs. The effective metric is therefore determined, up to the single temporal coefficient  $A_\tau$  (subject to Q5b-O3 [3]), by the representation-theoretic data already established in O23 [1], O28–O29 [2, 5], and Q7 [6].

## 7 Summary and Open Problems

### 7.1 What this paper establishes

1. The sub-principal  $k_Z^2$ -term in  $\sigma_2(L_{\text{eff}})$  has a unique algebraic source: the Heisenberg commutator  $[\tilde{X}, \tilde{Y}] = \tilde{Z}$  (Proposition 3.1).
2. Under the equivariant bridge, the coefficient of this term is fixed by the  $\mathfrak{su}(2)$ -Casimir:  $A_Z = C_{\mathfrak{su}(2)} = 2$  (Lemma 4.1, Proposition 4.2).
3. Consequently the effective spatial co-metric is isotropic with all eigenvalues equal to 2, and the full co-metric  $g^{\mu\nu}$  is non-degenerate rank-4 Lorentzian (Theorem 1.1, Corollary 6.1).
4. Open problem Q5b-O2 of [3] is resolved under the hypotheses of Theorem 1.1.

### 7.2 Remaining open problems

**Q8-O1** (*Bridge existence*). The proof of Q7 Conjecture 4.7 remains open. Theorem 1.1 establishes what the bridge must look like; existence itself is not proved here. This is the central open problem in the Q-series geometric emergence track.

**Q8-O2** (*Unconditional isotropy*). Q7 Conjecture 6.6 (asymptotic  $A_H \rightarrow 2$  as  $q \rightarrow \infty$ ) is supported by numerical data for  $q \in \{61, 101, 151\}$  but not proved analytically. An unconditional proof would remove the dependence on Q7 Conjecture 6.6 from Corollary 6.1. The expected difficulty is moderate: the power-law decay  $|A_H - A_Z| \sim q^{-0.44}$  suggests an approach via the asymptotic expansion of the Weil Laplacian spectrum.

**Q8-O3** (*Identification of  $A_\tau$  and completion of Q5b-O3*). With  $A_Z = 2$  established, the remaining free coefficient is  $A_\tau$  (the temporal sector of the effective metric). Its identification from the Born–Infeld saturation data is the subject of Q5b-O3 [3].

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